

	FIR VEHICLE TESTING		15/06/2023
FIR No :FIR/Testing-Outdoor/516900/2023/6117			
Part No/ Group :51695248R			
Subject :Performance evaluation of Signa 2821.T Cx RDE in comparison with Ashok Leyland 2820.T BSVI and Signa 2821.T BSVI base vehicle.			
Test Vehicle	Project Name :Signa 2821.T Cx 5LNGE (204 HP engine) OBDII BS-VI Vehicle Model Name : Signa 2821.T Cx 5LNGE		Enclosures : Annexure 1 - 6
	Chasis Number :PC03509022RJ00800 Engine Number :GXX103502		
FIR Status :	ESO FIR		

Role	Name	Signature
Prepared By	Palash Das	 15-Jun-2023
Checked By	Swapnil Jadhav	 15-Jun-2023
Approved By	Ajaykumar Jadhav	 15-Jun-2023

Cc: Head CVBU Engg, Chief Engineer, Head COC, Project Team [Design]; Project Team [Testing]

PA 20 - F 222 / May19

Proprietary and confidential information of TATA Motors Ltd. © Copyright TATA Motors Ltd.

Documents : Please click here

[FIR_Testing_Outdoor_51690..](#)

 FIR/Testing-Outdoor/516900/2023/6117		FIR VEHICLE TESTING		Date: 14/06/2023	
Part No / Group: 51695248R		Specification no :- " 259051690016 "		Action by	Failure class
Subject: -		Performance evaluation of Signa 2821.T Cx RDE in comparison with Ashok Leyland 2820.T BSVI and Signa 2821.T BSVI base vehicle.			
Conclusion:					
➤ <u>Signa 2821.T Cx RDE in comparison with Ashok Leyland 2820 HM BSVI</u> - Overall performance of Signa 2821.T Cx BSVI RDE is better as compared to Ashok Leyland 2820 BSVI HM PALS position "AL" achieved against "AL" ➤ <u>Signa 2821.T Cx RDE in comparison with Signa 2821.T BSVI</u> - Overall performance of Signa 2821.T Cx BSVI RDE is better as compared to Signa 2821.T BSVI base vehicle. 1.1 Acceleration: ➤ <u>Signa 2821.T Cx RDE in comparison with Ashok Leyland 2820 BSVI</u> - Acceleration (0-60 kmph) of Signa 2821.T Cx BSVI RDE was found to be 48.25 Sec. Signa 2821.T BSVI RDE is 4.22% better as compared to Ashok Leyland 2820 BSVI vehicle and meets PALS target. - Acceleration (0-1000m) of Signa 2821.T Cx BSVI RDE was found to be 75.3 Sec. Signa 2821.T Cx BSVI RDE is 0.71% better as compared to Ashok Leyland 2820 BSVI vehicle and meets PALS target. ➤ <u>Signa 2821.T Cx RDE in comparison with Signa 2821.T BSVI</u> - Acceleration (0-60 kmph) of Signa 2821.T Cx BSVI RDE is Superior by 4.92% as compared to Signa 2821.T BSVI vehicle. - Acceleration (0-1000m) of Signa 2821.T Cx BSVI RDE is Superior by 1.57% as compared to Signa 2821.T BSVI vehicle. 1.2 Wide open throttle drivability: ➤ <u>Signa 2821.T Cx RDE in comparison with Ashok Leyland 2820 BSVI</u> - In gear drivability of Signa 2821.T Cx BSVI RDE vehicle was found to be 2.25 kmph/s, 2.23 kmph/s, 1.83 kmph/s, 1.55 kmph/s, 1.07 kmph/s, and 0.61 kmph/s in 1st , 2nd , 3rd , 4th , 5th and 6th gear respectively.		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		
		--	--		

 FIR/Testing-Outdoor/516900/2023/6117		FIR VEHICLE TESTING		Date: 14/06/2023	
Part No / Group: 51695248R		Specification no :- " 259051690016 "		Action by	Failure class
Rate of acceleration of Signa 2821.T Cx BSVI RDE vehicle is superior to AL 2820.T BSVI in All gears except 1st Gear.				VI/ENG	C
1.3 <u>Bathtub drivability:</u>					
<u>Signa 2821.T RDE in comparison with Ashok Leyland 2820 BSVI</u>					
Bathtub drivability of Signa 2821.T Cx BSVI RDE vehicle was found better as compared to Ashok Leyland 2820.T BSVI vehicle in 3rd, 4th, 5th & 6th gear and meets PALS revised target.				--	--
<u>Signa 2821.T BSVI RDE in comparison with Signa 2821.T BSVI</u>					
Bathtub drivability of Signa 2821.T Cx BSVI RDE vehicle was found superior to Signa 2821.T BSVI vehicle in all gears.				--	--
1.4 <u>Sustain and Restart Gradability:</u>					
<u>Signa 2821.T Cx BSVI RDE in comparison with Ashok Leyland 2820 BSVI</u>					
Sustain gradeability of Signa 2821.T Cx BSVI RDE was found to be 28.33%. Signa 2821.T Cx BSVI RDE is 9.81% superior as compared to Ashok Leyland 2820.T BSVI vehicle.				--	--
Restart gradeability of Signa 2821.T Cx BSVI RDE was found to be 24.18%. Signa 2821.T Cx BSVI RDE is 13.41% superior as compared to Ashok Leyland 2820.T BSVI vehicle.				--	--
<u>Signa 2821.T Cx BSVI RDE in comparison with Signa 2821.T BSVI</u>					
Sustain gradeability of Signa 2821.T Cx BSVI RDE is 4.23% superior as compared to Signa 2821.T BSVI base vehicle.				--	--
Restart gradeability of Signa 2821.T Cx BSVI RDE is 2.28% superior as compared to Signa 2821.T BSVI base vehicle.				--	--
2.0 <u>Result:</u> Wide observation.					
FIR Status				Page 2 of 5	

 FIR/Testing-Outdoor/516900/2023/6117		FIR VEHICLE TESTING		Date: 14/06/2023		
Part No / Group: 51695248R		Specification no :- " 259051690016 "			Action by	Failure class

FAILURES CLASS:

- A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement.
- B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention.
- C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule.
- D: Customer perception issues and minor irritants.

3.0 Test Vehicle:

1. Signa 2821.T Cx BSVI RDE (ERC – 1248)
Chassis No. – PC03509022RJ00800
Engine No. – GXX103502

Enclosures:

- Annexure 1: Panel chart
Annexure 2: PAT Sheet
Annexure 3: PAT Summary
Annexure 4: Test request
Annexure 5: Test plan
Annexure 6: EOL

Input Doc Ref. : Testing-Outdoor/15048

DID No : NA

DDP No : NA

Work Order No. :- P.22.EA.501

Make :- Tata Motors Ltd

T I No. – 42197

Earlier Ref: -

- FIR/Testing-Outdoor/ 508500/2021/3980 Report on performance evaluation of Ashok Leyland 2820 HG BSVI vehicle in Comparison with Signa 2821.T Ashwamedh and Signa 2821.T 5L BSVI vehicle.

4.0 Project Status: DR3

5.0 Introduction:

- As a part of RDE program, Signa 2821.T Cx BSVI RDE vehicle was offered for performance evaluation, Ashok Leyland 2820 HM BSVI was taken as core benchmark and Signa 2821.T BSVI vehicle was taken as Base for performance comparison. This report will reveal detail of this trial with the status.

FIR Status

Page 3 of 5

 TATA FIR/Testing-Outdoor/516900/2023/6117		FIR VEHICLE TESTING	Date: 14/06/2023	
Part No / Group: 51695248R	Specification no :- " 259051690016 "		Action by	Failure class

6.0 Testing

➤ Following test was conducted during trials.

Sr. No.	Test Description	Standard Referred	Instrument used
1	Acceleration	TST_VT_WG_0002	VBOX SX with GPS, Weighbridge (CRS Gate)
2	WOT Drivability	TST_VT_WG_0002	
3	Bath -Tub Drivability	TST_VT_WG_0261	
4	Gradability	TST_VT_WG_11	
5	Restart Gradability	TST_VT_WG_127	
6	Coastdown	TST_VT_WG_36	

7.0 Vehicle details:

Sr. No.	Parameters		Base Vehicle : OBD-1	Core Benchmark	Test Vehicle : OBD-2
			SIGNA 2821.T BSVI	AL 2820.T BSVI	SIGNA 2821.T RDE BSVI
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)		ERC-1187	--	ERC-1248
2	Vehicle GVW/GCW	kg	28000	28000	28000
3	Engine Details (Make/cc/Norms/no. of Cyl.)		TML 5LNGE	H-Series CRS BSVI	TML 5LNGE
	Engine Max Power		200 HP @ 2200 rpm	200 hp @ 2400 rpm	204 HP @ 2200 rpm
	Engine Max Torque		850Nm @ 1000-1600 rpm	700 Nm @ 1200-1900 rpm	850Nm @ 1000-1600rpm
	Engine Calibration Maturity Level		As per production released	As per production released	95% Cal Maturity
4	Regeneration Strategy (Time ase/Soot base)		Time Based (110 Hrs)	--	Time Based (110 Hrs)
5	Gear Box (make/drive/FGR)		G950-FGR - 9.36,DD	ZF 6S-850 Gearbox	G950-FGR - 9.36,DD
6	Gear Ratios		1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00	1st: 8.51, 2nd: 4.66, 3rd: 2.73, 4th: 1.78, 5th: 1.27, 6th: 1, Rev:7.64	1-9.36 , 2-5.20 , 3-3.04 , 4-1.83 , 5-1.32 , 6-1.0
7	Tyre Details		4.88	5.57	5.12
8	All 295/90R 20 (JK Make)		F1 - Apollo 295/90R 20 16PR R1, R2 - Apollo 295/90R 20 16PR		
	LBSC	High & Low Engine Speed Breakpoint	RPM	--	--
		Vehicle Mass Threshold	kg	--	--
		Variable Acceleration 1, 2	kmph/sec	--	--
	VAM	Variable Acceleration Speed 1, 2	kmph	--	--
		GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph	--	--
	RSG	Maximum Accelerator Vehicle Speed	kmph	80	80
	Torque Factor	Torque Limit Medium Adjustment Factor	Balance	--	--
		Torque Limit Maximum Adjustment Factor	Economy	--	--
	PTP	Powertrain Protection (Enable / Disable)	Disable	Disable	Disable
		Torque Limit	Nm	NA	NA
		USFE / Torque Table Revision (1/2/3/n)	--	--	--
9	Fan Details (type/make/blades)		24" E- Viscous fan	--	24" E- Viscous fan
10	Base Cal Improvement (Cal Maturity %)		As per production released	As per production released	95% Cal Maturity

Vehicle Photographs			
----------------------------	---	--	---

FIR Status

Page 4 of 5

Format No. PA20 F222 (R4) March, 2021

Proprietary and confidential information of TATA Motors Ltd. ©Copyright TATA Motors Ltd.

 FIR/Testing-Outdoor/516900/2023/6117	FIR VEHICLE TESTING	Date: 14/06/2023		
Part No / Group: 51695248R	Specification no :- " 259051690016 "			Action by Failure class

8.0 Torque table:-

8.3.1 : Base vehicle (Signa 2821.T BS-VI OBD-I (200 HP))

FE Switch Torque Table (Torque Limit)										
	Load Gea	1st Gear	2nd Gear	3rd Gear	4th Gear	5th Gear	6th Gear	7th Gear	8th Gear	Reverse
Economy Mode	-	850	850	700	590	590	590	-	-	850
Balance Mode	-	850	850	700	700	700	700	-	-	850
Power Mode	-	850	850	850	850	850	850	-	-	850

8.3.2: Test vehicle (Signa 2821.T BS-VI OBD-II (204 HP))

FE Switch Torque Table (Torque Limit)										
	Load Gea	1st Gear	2nd Gear	3rd Gear	4th Gear	5th Gear	6th Gear	7th Gear	8th Gear	Reverse
Economy Mode	NA	850	850	700	590	590	590	-	-	850
Balance Mode	NA	850	850	700	700	700	700	-	-	850
Power Mode	NA	850	850	850	850	850	850	-	-	850

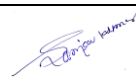
9.0 Observation:-

Overall performance Signa 2821.T Cx BSVI RDE is better as compared to Ashok Leyland 2820 HM BSVI and PALS position achieved "AL" against "AL".

10. Future testing: - As per further development.

FIR Status

Page 5 of 5

Level 1 Attribute:					PANEL chart Specification: Max Power-204 hp @ 2200 rpm Max Torque - 850 Nm 1000-1600 rpm Driveline : G-950 DD 5.12 RAR 295/90R 20 Tyres 3 Mode FE Switch Torque Factor: NA EOL Parameters : RSG: 80 kmph					VE CVBU/ PD/00		
PAT- Performance and Drivability										Panel Chart No.: VE CVBU/ M&HCV/ Signa 2821.T Cx BSVI RDE 5L/ PD/01		
Programme / MY : SIGNA 2821.T Cx 5L BSVI RDE			Gateway : DR3		Owner: Ajaykumar Jadhav			Project Name: SIGNA 2821.T 5L BSVI RDE(Cx Version)			Date: 14/06/2023	
Level 1 Summary:					Statement Of Intent (SOI) or DNA Description:							
PALS Status Current/ Gateway Job#1 Predicted PALS Level 1: AL AL					Performance: Haulage application							
Attribute Target Summary:												
Deriv.	Level 2	Level 3	Level 4	Level 5	Level 6	PALS	Metric	Current Target	Status	Base Vehicle	Competitor Data	Remarks
								w.r.t Ashok Leyland 2820.T BSVI	Current/Gateway SIGNA 2821.T Cx BSVI RDE	Signa 2821.T BSVI	Core Benchmark Ashok Leyland 2820.T	
									Heavy mode	Heavy mode	--	
	Specification								--	--	--	--
	Vehicle GVW (Payload)						kg		28000	28000	28000	
	Power (Max) @ rpm						HP		204 hp @ 2200 rpm	200 hp @ 2200 rpm	200 hp @ 2400 RPM	
	Power to Weight ratio						HP / Ton		7.29	7.14	7.14	
	Torque (Max) @ rpm						Nm		850 Nm @ 1000-1600 rpm	850 Nm @ 1000-1600 rpm	700 Nm at 1200-2000 rpm	
	Torque to Weight ratio						Nm / Ton		30.36	30.36	25.00	
	Tyres size						~		Bridgestone 295/90R20	JK 295/90 R20	Apollo JK Tyres 295/90 R20	
	Gear Box Ratios						~		G950 DD, FGR -9.36	G950 DD, FGR -9.36	Eaton DD 6S Gearbox	
	Final Drive Ratio						~		5.12	4.88	5.57	
	Acceleration at rated load					AL			AL			
	0-60 kmph						Sec	≤ 50.43	48.30	50.8	50.43	
	0-1000 m						Sec	≤ 75.84	75.30	76.50	75.84	
	Bathtub Drivability (Time) at Rated load					AL			AL			
	3rd Gear											
	15 - 20 Kmph						sec	≤ 3.24	2.22	3.06	3.24	
	20 - 25 Kmph						sec	≤ 2.48	2.10	2.85	2.48	
	25 - 30 Kmph						sec	≤ 2.62	--	3.21	2.62	
	4th Gear											
	20 - 25 Kmph						sec	≤ 4.55	4.53	5.31	4.55	
	25 - 30 Kmph						sec	≤ 4.33	3.19	4.19	4.33	
	30 - 35 Kmph						sec	≤ 3.89	3.21	4.20	3.89	
	5th Gear											
	30-40 kmph						sec	≤ 12.83	10.12	11.97	12.83	
	40-50 kmph						sec	≤ 12.74	9.85	12.24	12.74	
	50-60 kmph						sec	≤ 14.14	11.86	13.72	14.14	
	6th Gear											
	30-40 kmph						sec	≤ 17.44	25.28	--	17.44	
	40-50 kmph						sec	≤ 17.68	14.12	16.85	17.68	
	50-60 kmph						sec	≤ 17.77	14.02	16.87	17.77	
	60-70 kmph						sec	≤ 20.12	16.44	18.25	20.12	
	Gradeability at rated load					AL			AL			
	Sustain Gradeability						%	≥ 25.80	28.33	27.18	25.80	
	Restart Gradeability						%	≥ 21.32	24.18	23.64	21.32	
Executive Summary:		1. Ashok Leyland 2820.T BSVI have been taken for target setting. 2. Signa 2821.T Cx BSVI RDE was having 95% engine calibration maturation. 3.Target has been taken at-par as per Cx variant approach										
Tested by:								Verified and Approved by:				
 Palash Das / Swapnil Jadhav PAT Engineer								  Ajaykumar Jadhav / Sanjeev Kumar PAT Lead / Project Lead				

OUR CULTURE CREDO

AT TATA MOTORS

We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.

We are **MORE WHEN ONE!**

PED evaluation of Signa 2821.T RDE BSVI in comparison with Ashok Leyland 2820 BSVI and Signa 2821.T Ashwamedh BSVI

PED| 14.06.23



- ✓ Vehicle specification comparison
- ✓ Performance Data Summary
- ✓ Minimum and Maximum Gear Speed
- ✓ Wide open throttle performance
- ✓ Tractive effort comparison
- ✓ Acceleration
- ✓ WOT drivability
- ✓ Bathtub drivability
- ✓ Sustain and restart Gradeability

Vehicle Specification Comparison

Sr. No.	Parameters			Base Vehicle : OBD-1	Core Benchmark	Test Vehicle : OBD-2
				SIGNA 2821.T BSVI	AL 2820.T BSVI	SIGNA 2821.T RDE BSVI
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)			ERC-1187	--	ERC-1248
2	Vehicle GVW/GCW	kg		28000	28000	28000
3	Engine Details (Make/cc/Norms/no. of Cyl.)			TML 5LNGE	H-Series CRS BSVI	TML 5LNGE
	Engine Max Power			200 HP @ 2200 rpm	200 hp @ 2400 rpm	204 HP @ 2200 rpm
	Engine Max Torque			850Nm @ 1000-1600 rpm	700 Nm @ 1200-1900 rpm	850Nm @ 1000-1600rpm
	Engine Calibration Maturity Level			As per production released	As per production released	95% Cal Maturity
4	Regeneration Strategy (Time ase/Soot base)			Time Based (110 Hrs)	--	Time Based (110 Hrs)
5	Gear Box (make/drive/FGR)			G950-FGR - 9.36,DD	ZF 6S-850 Gearbox	G950-FGR - 9.36,DD
	Gear Ratios			1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00	1st: 8.51, 2nd: 4.66, 3rd: 2.73, 4th: 1.78, 5th: 1.27, 6th: 1, Rev:7.64	1-9.36 , 2-5.20 , 3-3.04 , 4-1.83 , 5-1.32 , 6-1.0
6	RAR			4.88	5.57	5.12
7	Tyre Details			All 295/90R 20 (JK Make)	F1 - Apollo 295/90R 20 16PR R1, R2 - Apollo 295/90R 20 16PR	All 295/90R 20 (Bridgestone Make)
8	LBSC	High & Low Engine Speed Breakpoint	RPM	--	--	--
		Vehicle Mass Threshold	kg	--	--	--
	VAM	Variable Acceleration 1, 2	kmph/sec	--	--	--
		Variable Acceleration Speed 1, 2	kmph	--	--	--
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph	--	--	--
	RSG	Maximum Accelerator Vehicle Speed	kmph	80	80	80
	Torque Factor	Torque Limit Medium Adjustment Factor	Balance	--	--	--
		Torque Limit Maximum Adjustment Factor	Economy	--	--	--
	PTP	Powertrain Protection (Enable / Disable)		Disable	Disable	Disable
		Torque Limit	Nm	NA	NA	NA
	USFE / Torque Table Revision (1/2/3/n)			--	--	--
9	Fan Details (type/make/blades)			24" E- Viscous fan	--	24" E- Viscous fan
10	Base Cal Improvement (Cal Maturity %)			As per production released	As per production released	95% Cal Maturity

Performance Data Summary

Parameter	Unit			Benchmark	Base Vehicle	Test Vehicle	RDE Status (%) w.r.t Target	RDE Status (%) w.r.t Base	Remarks	
		PALS Position	Target 2821.T RDE	Ashok Leyland 2820 HG BSVI Eaton 6S DD RAR 5.57	Signal 2821.T BSVI_HM RAR 4.88	Signal 2821.T 5L RDE BSVI_HM RAR 5.12				
Acceleration										
0 - 60 kmph	sec	C	≤50.43	50.43	50.8	48.3	4.22%	4.92%		
0 - 1000 m	sec	C	≤75.84	75.84	76.50	75.3	0.71%	1.57%		
Driveability										
1st Gear(4-6)	sec		Not Applicable	0.84	1.02	0.89	Target Not available	Target Not available		
2nd Gear(8-12)	sec			2.32	1.99	1.79				
3rd Gear(12-20)	sec			6.04	5.04	4.36				
4th Gear(19-31)	sec			5.69	5.40	4.52				
5th Gear(25-42)	sec			11.00	9.60	8.45				
6th Gear(36-50)	sec			23.40	24.86	22.86				
Rate of acceleration										
1st Gear(4-6)	kmph/sec		Not Applicable	2.38	1.96	2.25	Target Not available	Target Not available		
2nd Gear(8-12)	kmph/sec			1.72	2.01	2.23				
3rd Gear(12-20)	kmph/sec			1.32	1.59	1.83				
4th Gear(19-31)	kmph/sec			1.23	1.30	1.55				
5th Gear(25-42)	kmph/sec			0.82	0.94	1.07				
6th Gear(36-50)	kmph/sec			0.60	0.56	0.61				
Bath-Tub Driveability										
3rd Gear										
15-20	Sec	C	≤ 3.24	3.24	3.06	2.22	31.64%	27.61%		
20-25	Sec	C	≤ 2.48	2.48	2.85	2.10	15.32%	26.32%		
25-30	Sec	C	≤ 2.62	2.62	3.21					
4th Gear										
20-25	Sec	C	≤ 4.55	4.55	5.31	4.53	0.44%	14.69%		
25-30	Sec	C	≤ 4.33	4.33	4.19	3.19	26.33%	23.87%		
30-35	Sec	C	≤ 3.89	3.89	4.20	3.21	17.48%	23.57%		
5th Gear										
30-40	Sec	C	≤ 12.83	12.83	11.97	10.12	21.10%	15.43%		
40-50	Sec	C	≤ 12.74	12.74	12.24	9.85	22.67%	19.51%		
50-60	Sec	C	≤ 14.14	14.14	13.72	11.86	16.11%	13.54%		
6th Gear										
30-40	Sec	C	≤ 17.44	17.44		25.28	-44.95%			
40-50	Sec	C	≤ 17.68	17.68	16.85	14.12	20.14%	16.20%		
50-60	Sec	C	≤ 17.77	17.77	16.87	14.02	21.09%	16.88%		
60-70	Sec	C	≤ 20.12	20.12	18.25	16.44	18.29%	9.92%		
Gradability and Restartability										
Sustain Gradability	%	C	≥ 25.8	25.80	27.18	28.33	9.81%	4.23%		
Restart Gradability	%	C	≥ 21.32	21.32	23.64	24.18	13.41%	2.28%		

Signal 2821.T Cx BSVI RDE Performance. PALS position achieved "AL" against "AL"

Acceleration

- Acceleration of Signal 2821.T Cx BSVI RDE was found to be **better as compared to target** in 0-60 kmph speed and 0-1000 meters band

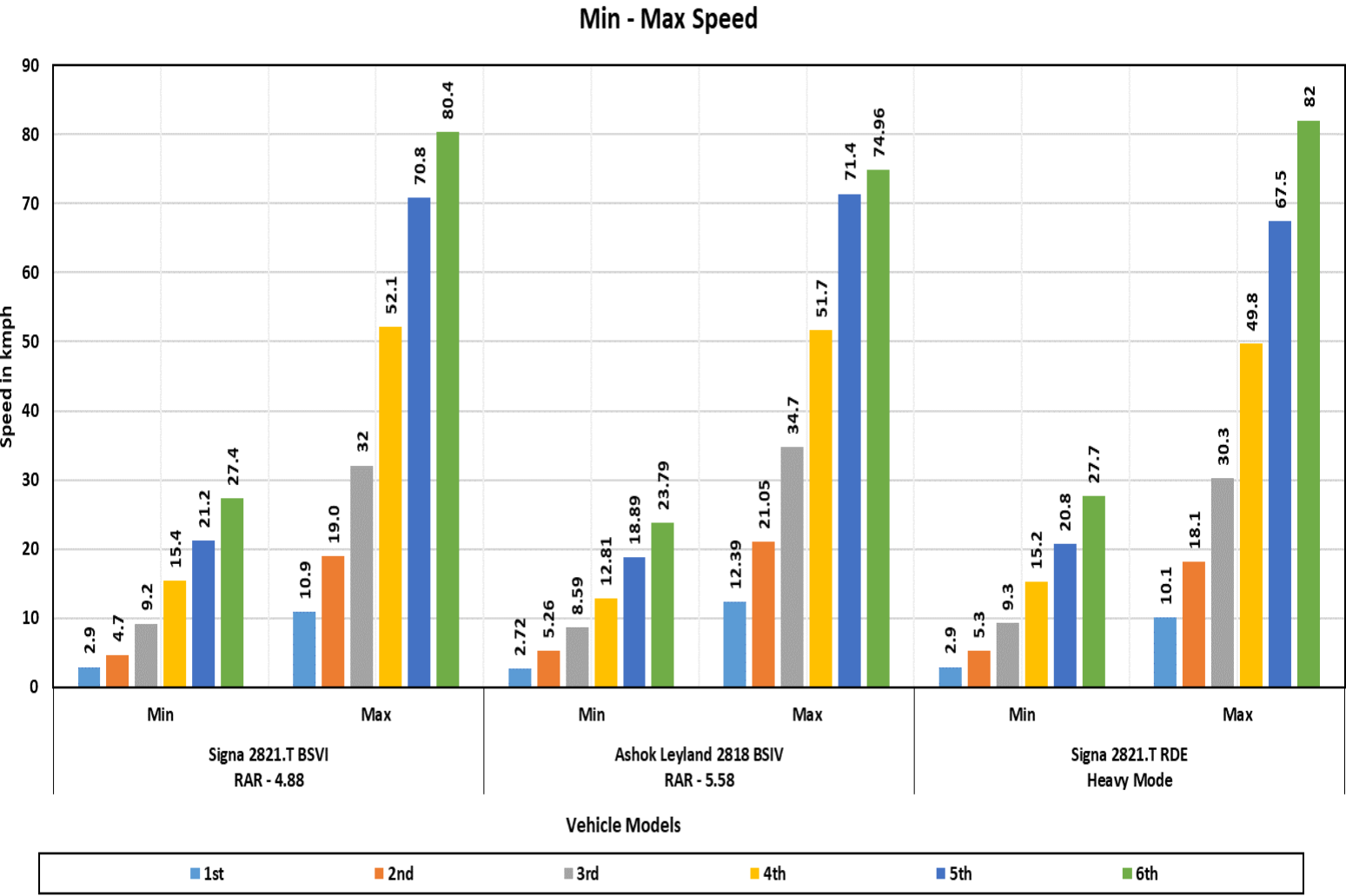
Bath tub Drivability

- Drivability of Signal 2821.T Cx BSVI RDE was found to be **better as compared to target** in all the speed bands except 6th gear 30-40 speed band

Sustain & Restart Gradeability

- Restart Gradability of 2821.T Cx RDE BSVI was **found better** than target.
- Sustain Gradability of 2821.T Cx RDE BSVI was **found better** than target

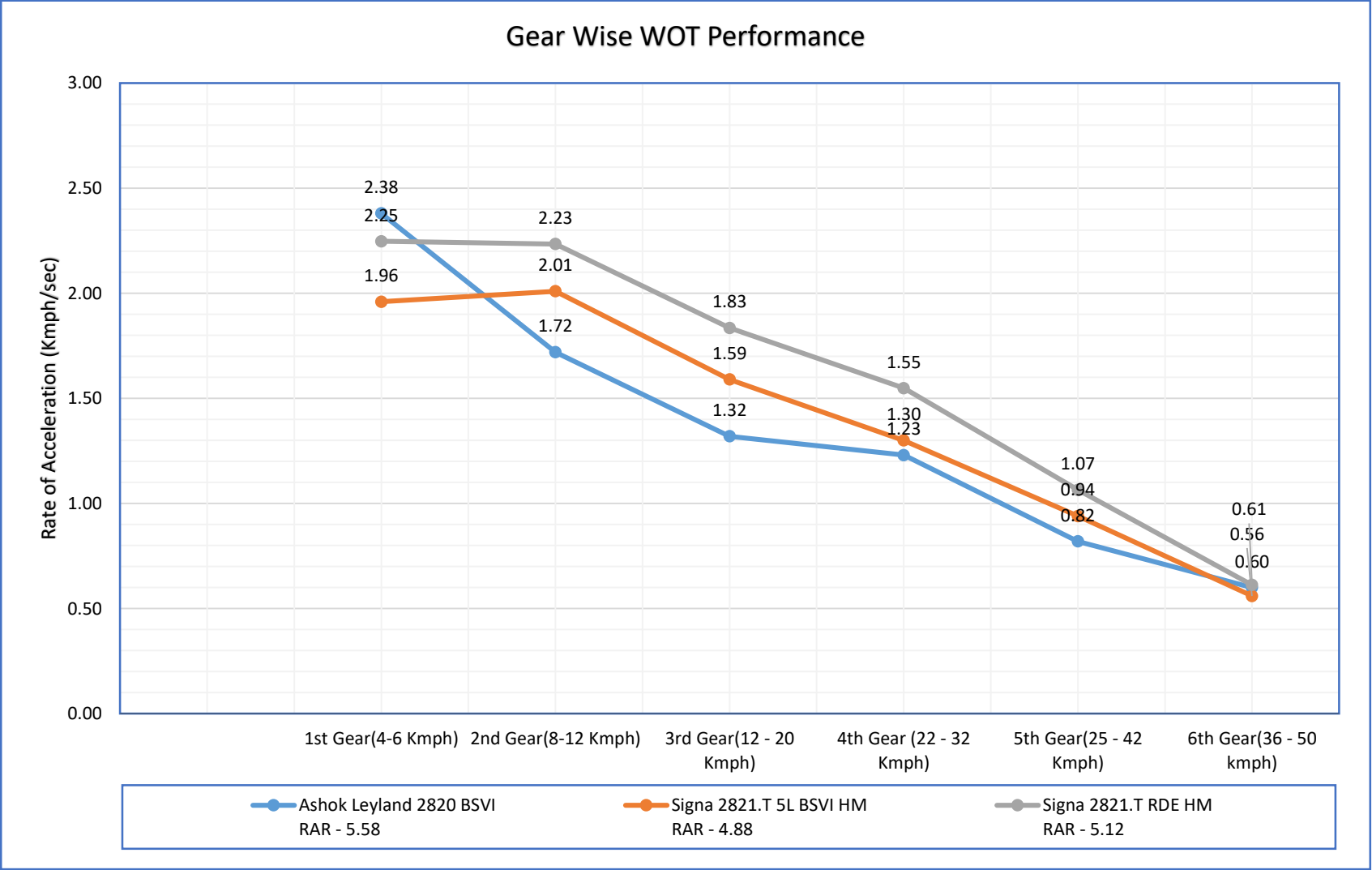
Minimum & Maximum Gear Wise speed Comparison



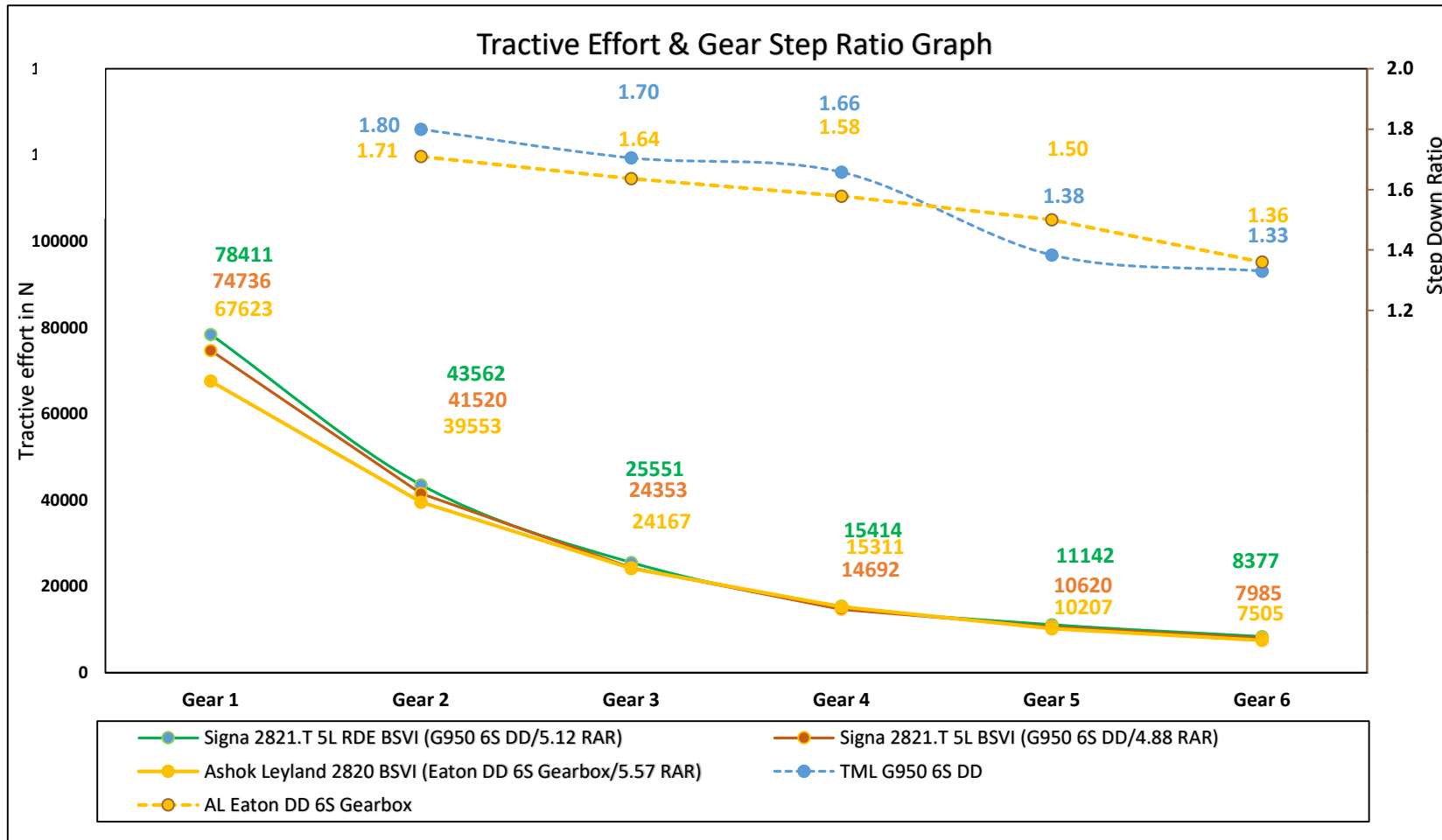
Gear	Signa 2821.T BSVI RAR - 4.88		Ashok Leyland 2818 BSIV RAR - 5.58		Signa 2821.T RDE Heavy Mode	
	Min	Max	Min	Max	Min	Max
1st	2.9	10.9	2.72	12.39	2.9	10.1
2nd	4.7	19.0	5.26	21.05	5.3	18.1
3rd	9.2	32	8.59	34.7	9.3	30.3
4th	15.4	52.1	12.81	51.7	15.2	49.8
5th	21.2	70.8	18.89	71.4	20.8	67.5
6th	27.4	80.4	23.79	74.96	27.7	82

Wide open throttle Performance

Gear	Ashok Leyland 2820 BSVI RAR - 5.58	Signal 2821.T 5L BSVI HM RAR - 4.88	Signal 2821.T RDE HM RAR - 5.12
1st Gear(4-6 Kmph)	2.38	1.96	2.25
2nd Gear(8-12 Kmph)	1.72	2.01	2.23
3rd Gear(12 - 20 Kmph)	1.32	1.59	1.83
4th Gear (22 - 32 Kmph)	1.23	1.30	1.55
5th Gear(25 - 42 Kmph)	0.82	0.94	1.07
6th Gear(36 - 50 kmph)	0.60	0.56	0.61



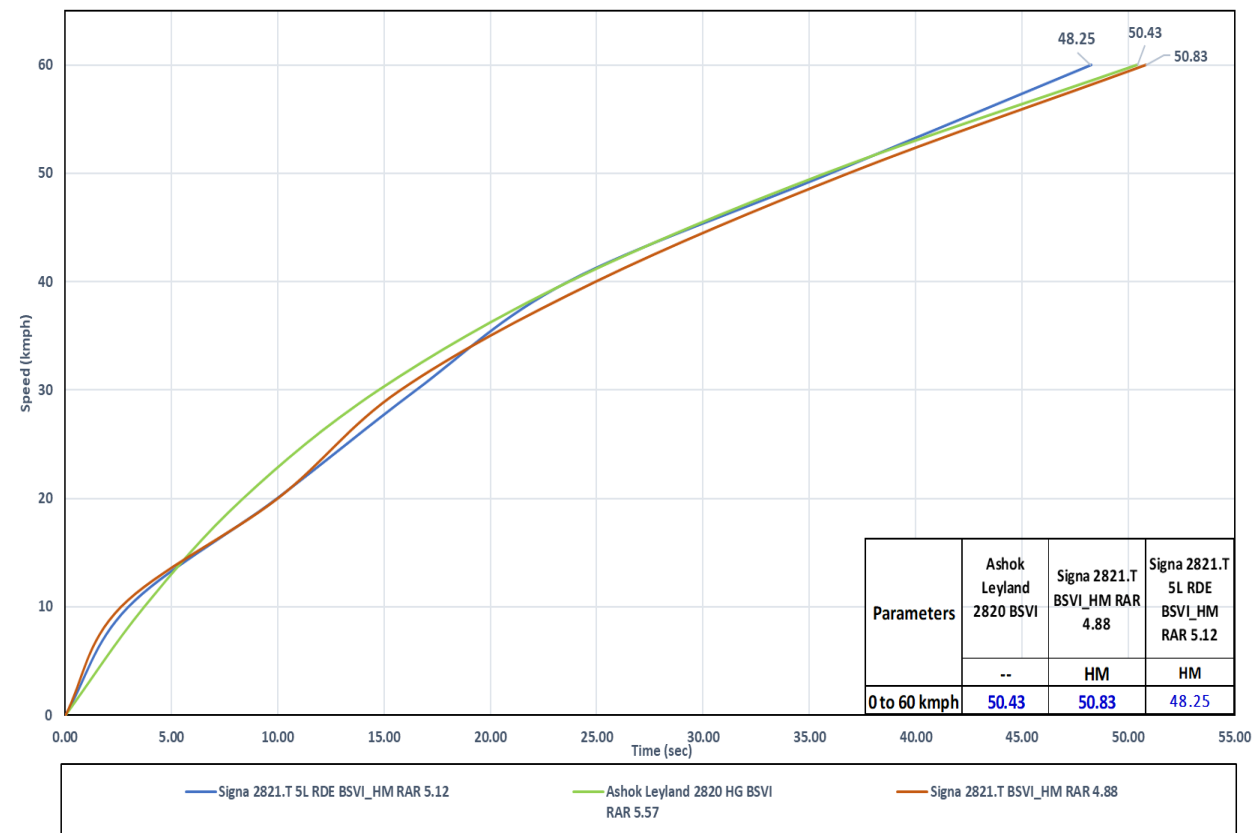
Tractive effort



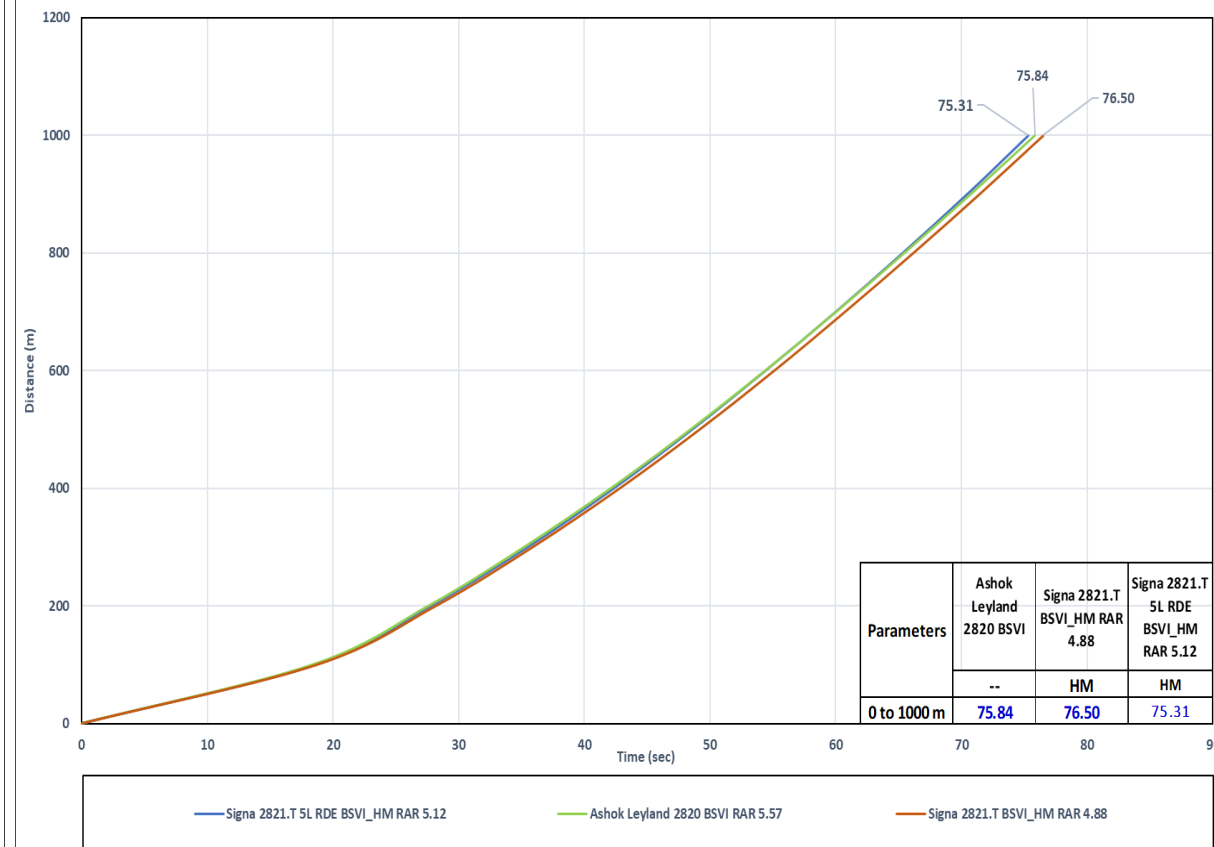
❖ Tractive effort of Signa 2821.T Cx RDE vehicle **is better as compared** with AL 2820.T BSVI & Signa 2821.T BSVI in all gear.

Acceleration Performance – Through Gear

Acceleration 0 to 60kmph



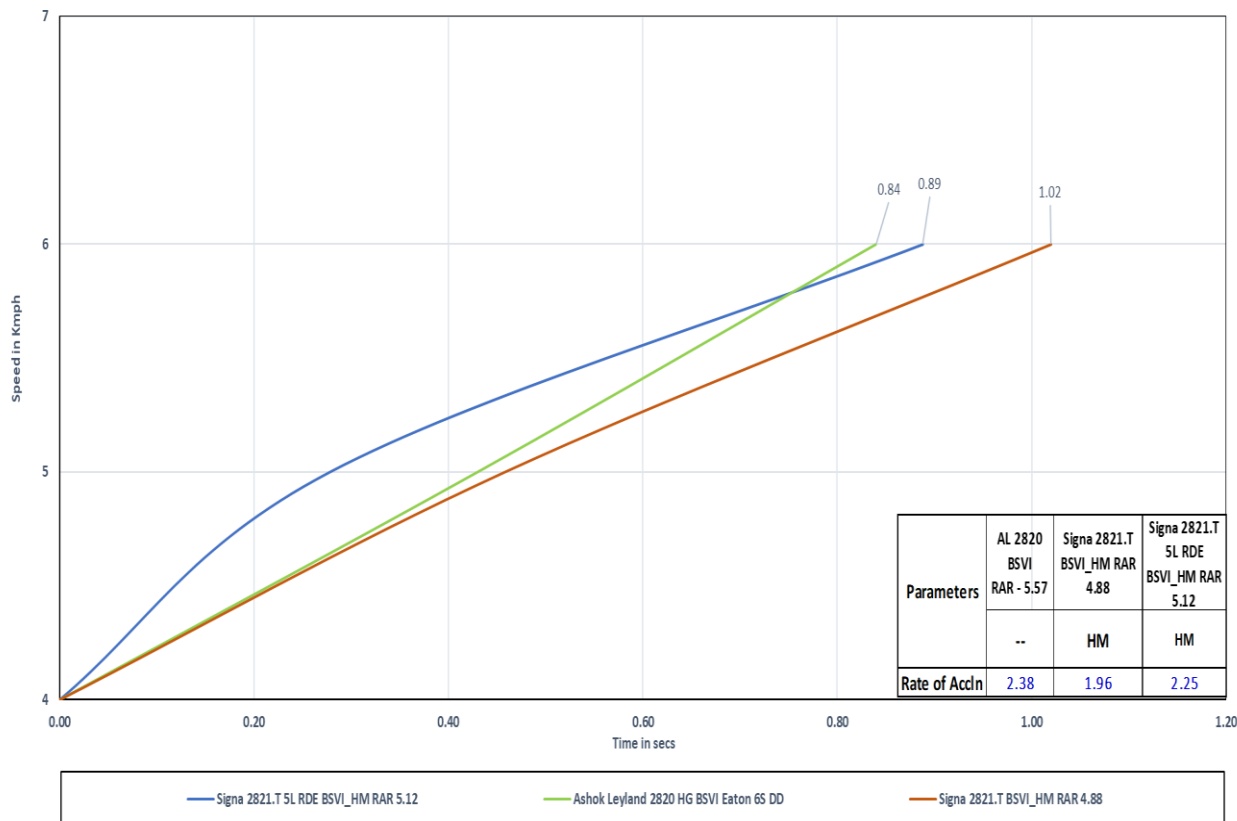
Acceleration 0 to 1000m



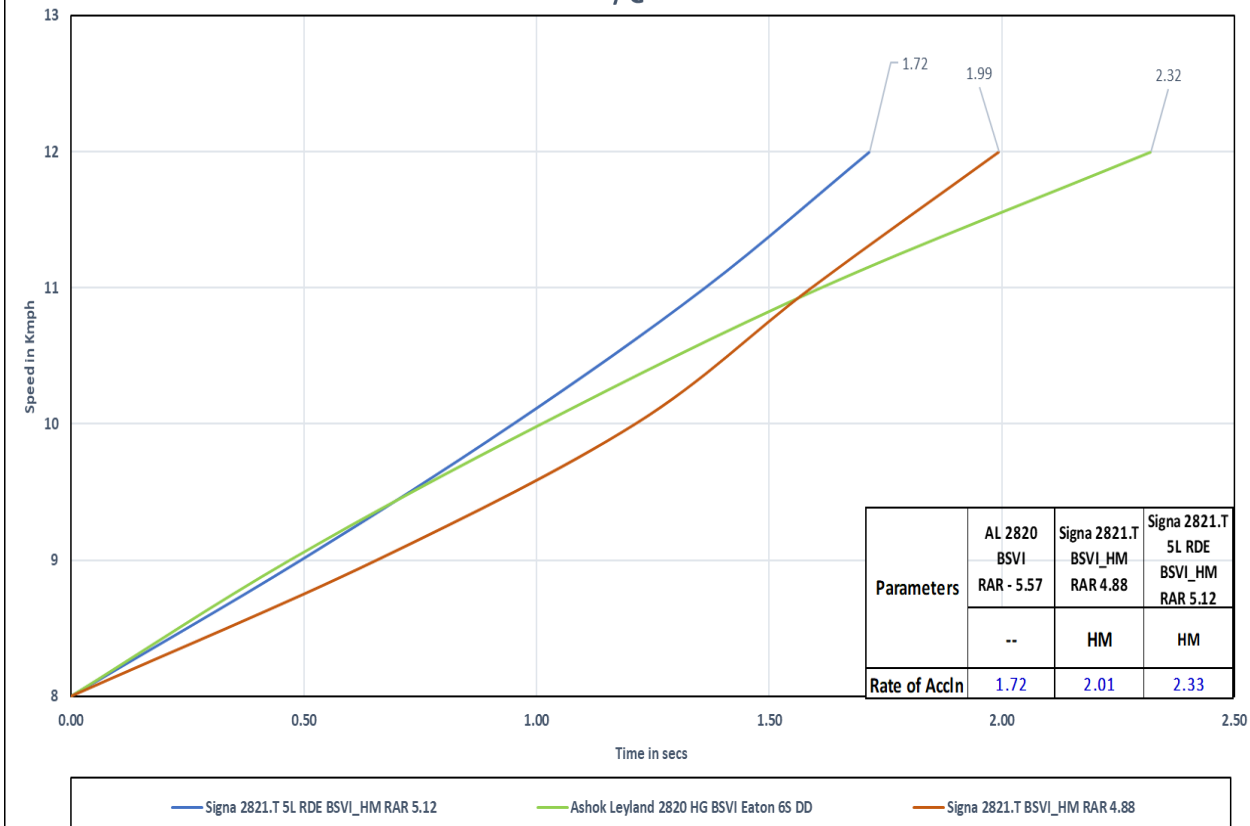
❖ Acceleration (0-60kmph) & (0-1000M) , Signa 2821.T Cx RDE vehicle **is better as compared** with AL 2820.T BSVI & Signa 2821.T BSVI

WOT Drivability @ 1st & 2nd Gear

Drivability @ 1st Gear



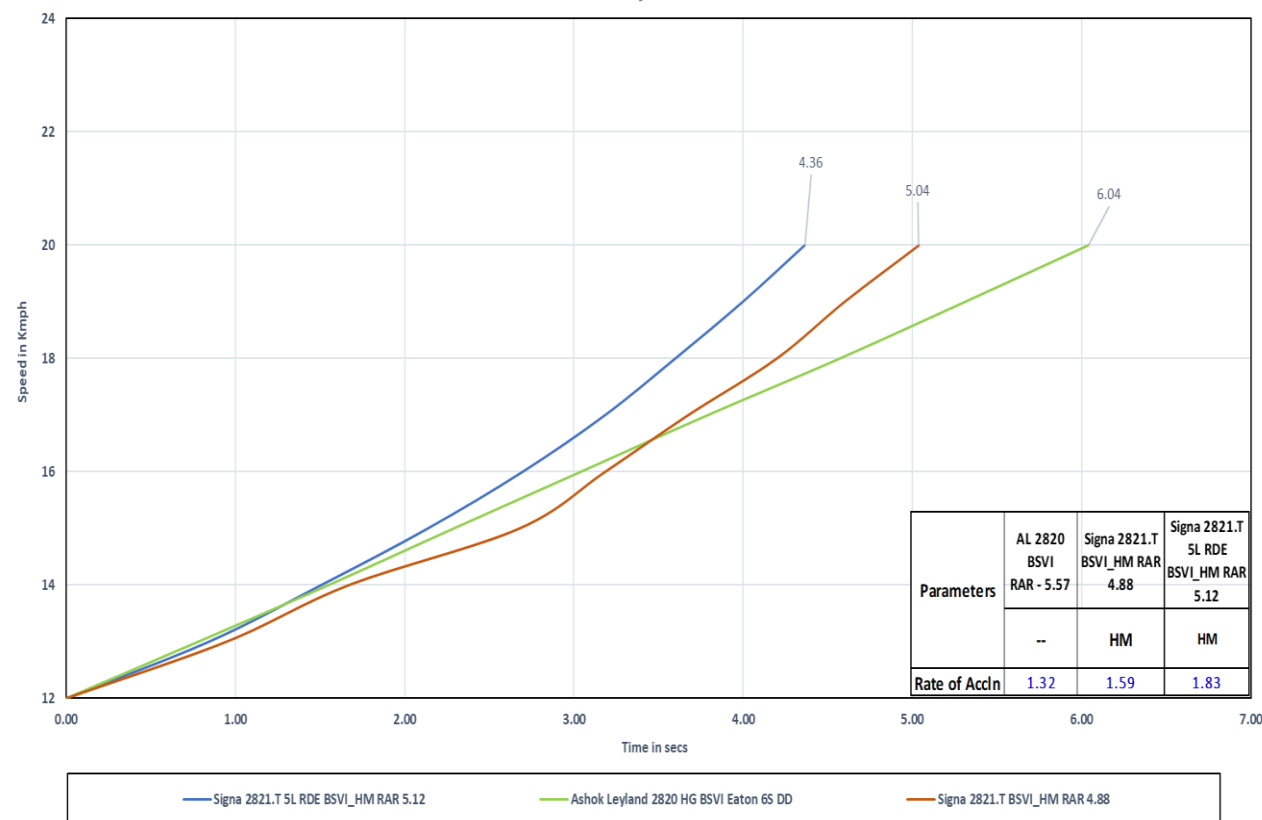
Drivability @ 2nd Gear



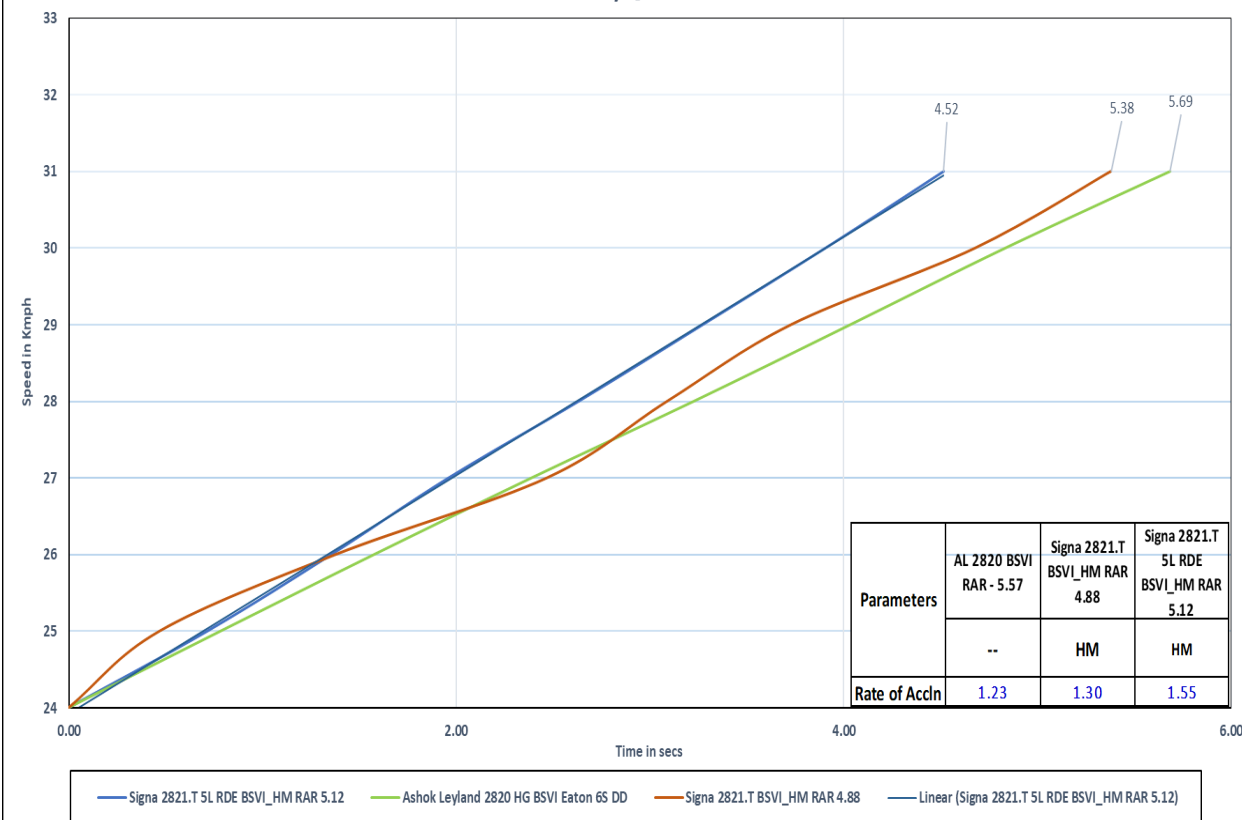
❖ In Second gear, WOT acceleration performance of Signa 2821.T Cx RDE vehicle **is better as compared** with AL 2820.T BSVI & Signa 2821.T BSVI

WOT Drivability @ 3rd & 4th Gear

Drivability @ 3rd Gear

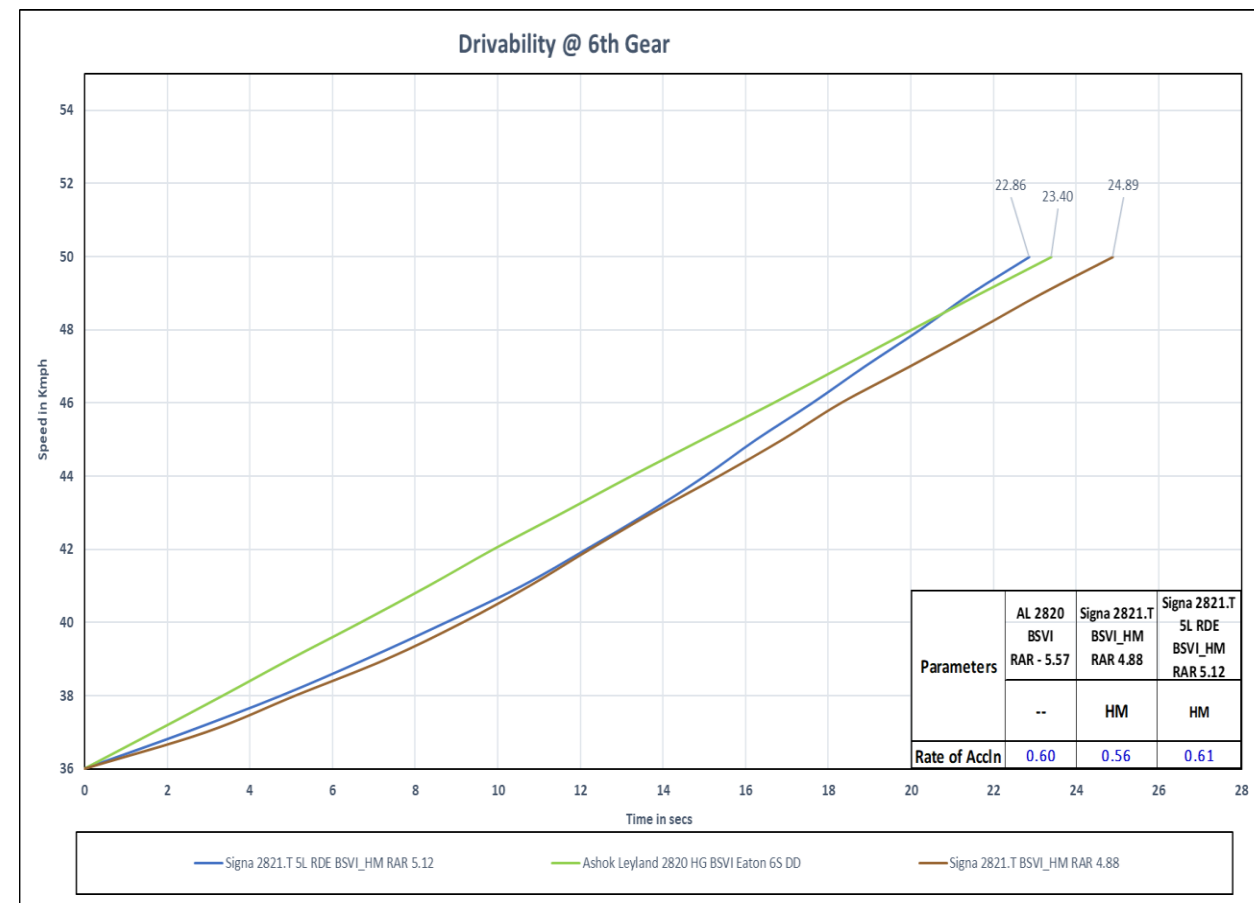
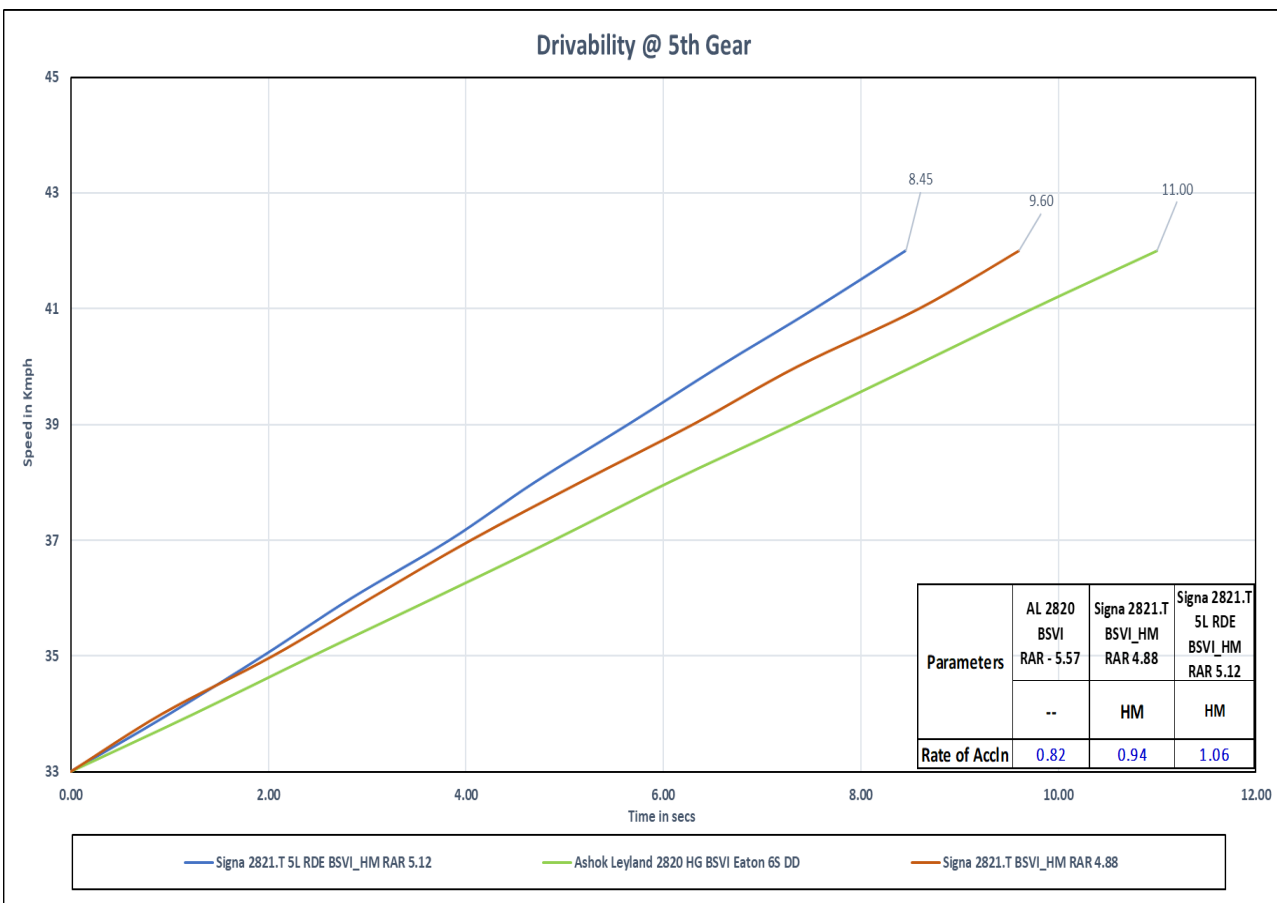


Drivability @ 4th Gear



- ❖ In third gear, WOT acceleration performance of Signa 2821.T Cx RDE vehicle **is better as compared** with AL 2820.T BSVI & Signa 2821.T BSVI
- ❖ In fourth gear, WOT acceleration performance of Signa 2821.T Cx RDE vehicle **is better as compared** with AL 2820.T BSVI & Signa 2821.T BSVI

WOT Drivability @ 5th & 6th Gear

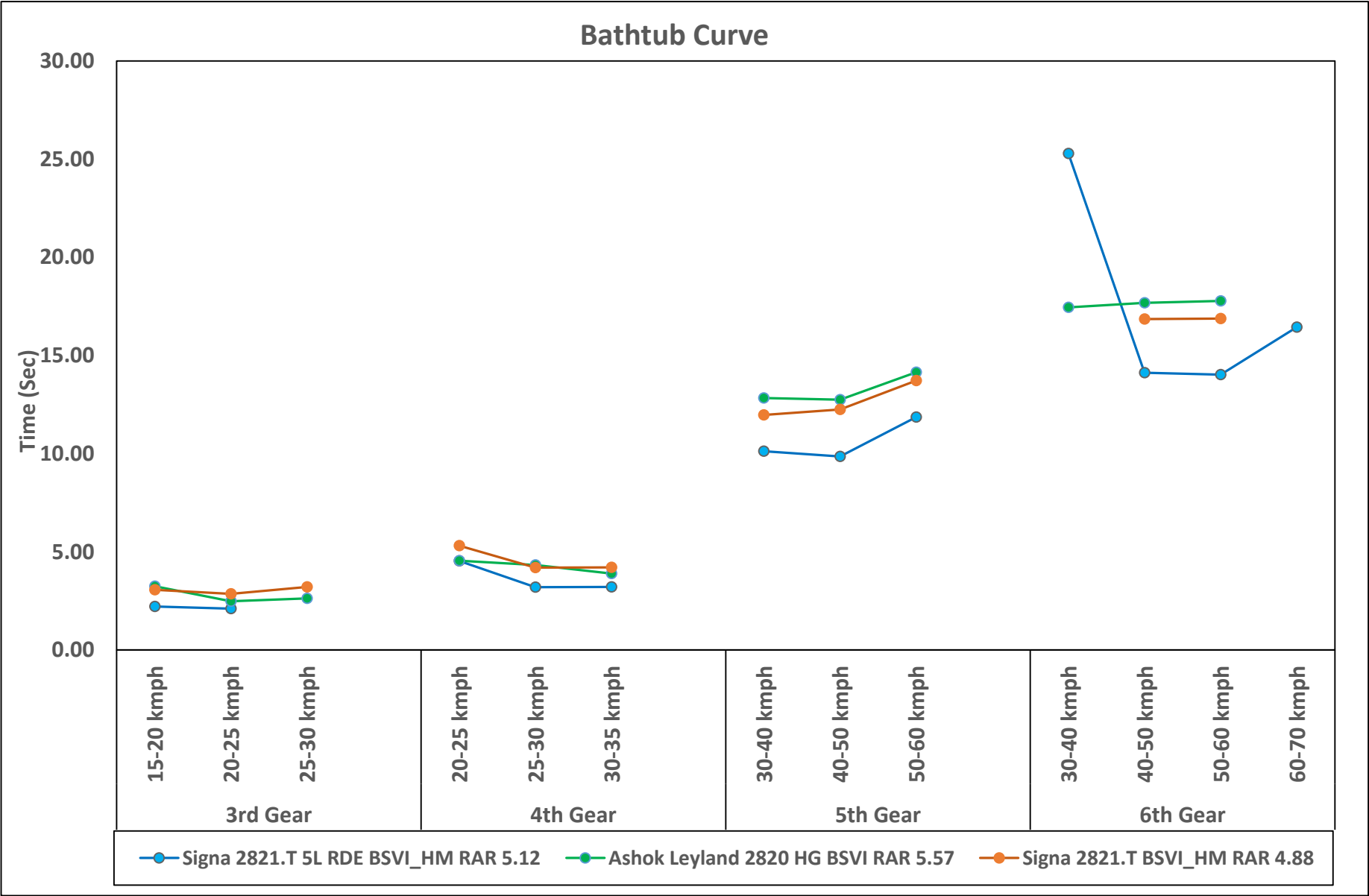


- ❖ In Fifth gear, WOT acceleration performance of Signa 2821.T Cx RDE vehicle **is better as compared** with AL 2820.T BSVI & Base vehicle
- ❖ In Sixth gear, WOT acceleration performance of Signa 2821.T Cx RDE vehicle **is better as compared** with AL 2820.T BSVI

Bathtub Drivability Report

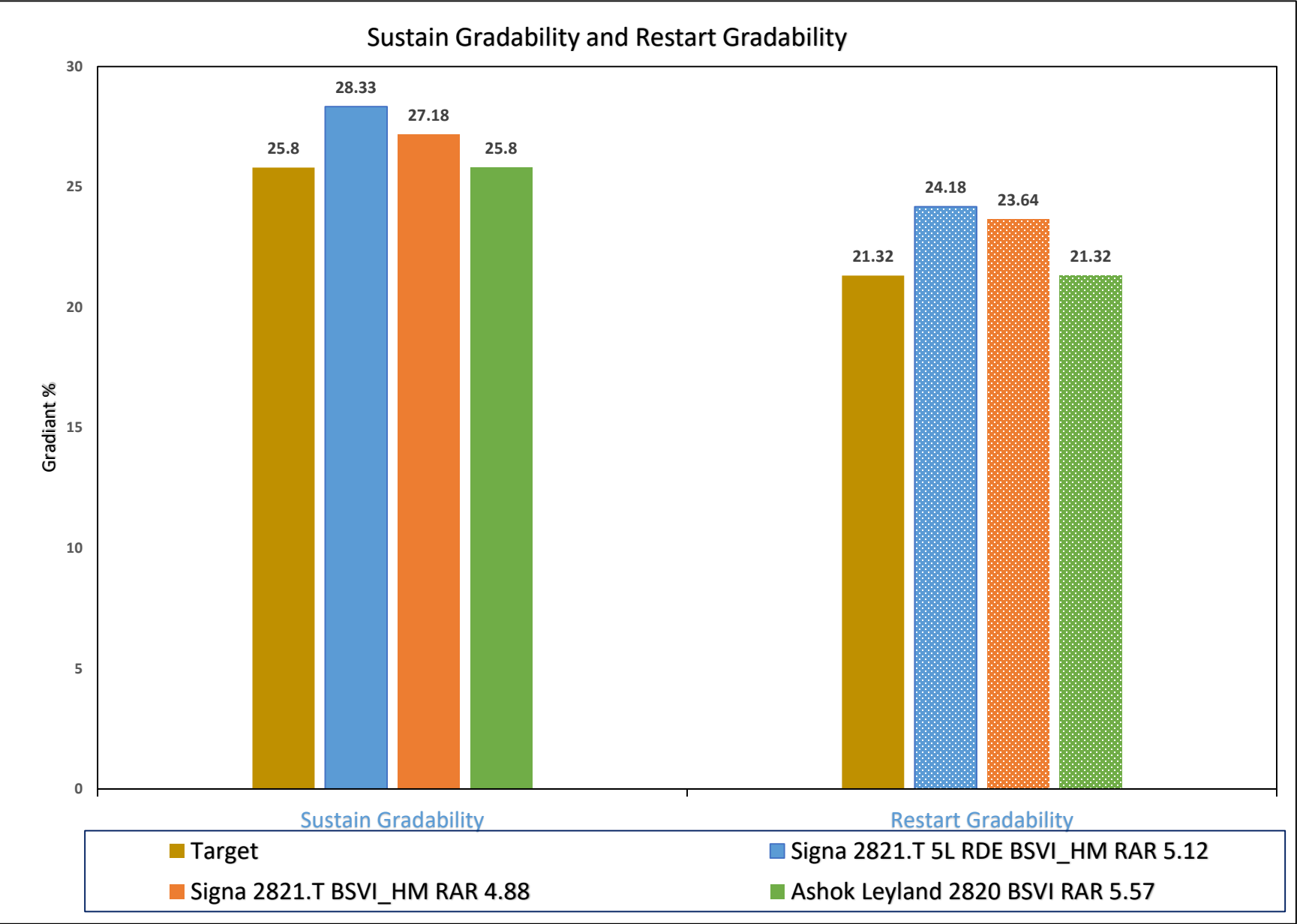
Gear	Speed band	Ashok Leyland 2820 HG BSVI RAR 5.57	Signa 2821.T BSVI_HM RAR 4.88	Signa 2821.T 5L RDE BSVI_LM RAR 5.12	Signa 2821.T 5L RDE BSVI_BM RAR 5.12	Signa 2821.T 5L RDE BSVI_HM RAR 5.12
		Time in Sec	Time in Sec	Time in Sec	Time in Sec	Time in Sec
3rd Gear	15-20 kmph	3.24	3.06	2.46	2.46	2.22
	20-25 kmph	2.48	2.85	2.34	2.34	2.10
	25-30 kmph	2.62	3.21			
4th Gear	20-25 kmph	4.55	5.31	5.09	4.78	4.53
	25-30 kmph	4.33	4.19	4.80	3.93	3.19
	30-35 kmph	3.89	4.20	4.71	3.91	3.21
5th Gear	30-40 kmph	12.83	11.97	14.43	12.07	10.12
	40-50 kmph	12.74	12.24	14.92	12.14	9.85
	50-60 kmph	14.14	13.72	15.00	13.38	11.86
6th Gear	30-40 kmph	17.44		26.54	26.02	25.28
	40-50 kmph	17.68	16.85	22.70	17.26	14.12
	50-60 kmph	17.77	16.87	23.88	18.46	14.02
	60-70 kmph	20.12	18.25		19.97	16.44

Bathtub Drivability Comparison with Benchmark & Base Vehicles



Remarks:
Vehicle is meeting PALS target
in all gears

Sustained and Restart Gradeability



Test	Sustain Gradability and Restart Gradability			
	Target	Ashok Leyland 2820 BSVI RAR 5.57	Signa 2821.T BSVI_HM RAR 4.88	Signa 2821.T 5L RDE BSVI_HM RAR 5.12
Sustain Gradability	25.8	25.8	27.18	28.33
Restart Gradability	21.32	21.32	23.64	24.18

Thank You!

TEST REQUISITION

Ref.		Date:	31/03/2023
------	--	-------	------------

To,

ERC Testing

Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 2821.T 5L BSVI				
2	Part Number	51695248R,				
3	Drawing number	516900000003,	Mod status	B,2,	Mod Date	25/02/2023
4	Vehicle Model / Project	TATA SIGNA 2821.T RDe BSVI				
5	WBS Number	P.22.EA.501 -RDE Truck 5.6 Ltr 3123.T / 3523.T				
6	Supplier	NA				
7	Number of samples					
8	Inspection report	Enclosed	☒ Yes	☐ No	If No commitment date:	
9	Material report	Enclosed	☐ Yes	☒ No	If No commitment date:	
10	Suppliers test report	Enclosed	☒ Yes	☐ No	If No commitment date:	
11	Reason for testing	First Source Development	☐	Weight Reduction	☐	
		Alternate source development	☐	Cost Reduction	☐	
		Data generation Test	☐	Regulatory test	☐	
		Any other (pl. specify)	☐	New Development	☐	
12	What testing required?	DYNAMIC				
13	Remarks					
14	Peripheral Parts Availability	YES				

Authorized Signature:	
Name:	
Dept:	

Contact Person:	
Phone Number	

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

TATA MOTORS LIMITED ERC PUNE	TEST PLAN		DATE: 14/06/2023				
	VEHICLE TESTING DEPARTMENT						
Description: TATA SIGNA 2821.T RDE BSVI			Test Item No: 42197				
			New	✓	Moderate	✗	
			Major	✗	Minor	✗	

Part No: 51695248R,	Modification No: B,2,	Enclosures: PD Trials of Signa 2821.T 5L RDE BSVI Cx Vehicle
Drawing No: 5169000000003,		
Make: NA	WBS Number: P.22.EA.501 -RDE Truck 5.6 Ltr 3123.T / 3523.T	Copies to :
Vehicle Type: TATA SIGNA 2821.T RDe BSVI	No. of Samples: 1	
Test Requested by:	VEHICLE-I	Equipments:
Test Requisition Reference:	Testing-Outdoor/15048	
Reason for Test/s:	Component & System Level Performance Tests	
Regulatory Requirement:	a) Component:	
	b) Vehicle:	

Sr. No.	Test description	Test specification	Acceptance criteria	No. of Samples	Test Responsibility	Proto Phase (Mule/ Alpha/ Beta/PP)	Remarks
1	TST/VT/ WG/158	Performance driveability	yes	1	Palash Das	Beta	completed

REMARKS: Performance driveability	INTERNAL REF:	
TEST TO FAILURE: No		
MEOST: No		
RELIABILITY TESTING: No		
BENCHMARKING TESTING: No		
DFMEA/FMEA DONE: Yes		
PREPARED BY	CHECKED BY	APPROVED BY